

Privacy in Machine Learning (Attacks)

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High Level Overview

★ The Secret Sharer (2019)

- Use “auto-complete” queries to attack privacy of out-of-distribution data trained on by NLP generative models
- Suggests Differential Privacy (DP) as a solution

★ Deep Models Under the GAN (2017)

- Use generative models (GAN) to attack “record-level” Differential Privacy (DP) as used in Federated Learning (FL) setting
- Suggests “user-level” Differential Privacy (DP) as a solution

The Secret Sharer (2019)

The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks

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Paper Overview

- ★ Why is unintended memorization a problem?
 - Demonstrate potential “auto-complete” privacy attack
 - Define exposure metric to quantify unintended memorization
 - Run experiments proving existence of attack according to metric
- ★ How effective is exposure metric?
 - Define extraction algorithm as an effective and efficient attack strategy
 - High exposure ☐ Extraction
- ★ What are potential defenses to this attack?
 - Differential privacy

Definitions: Unintended Memorization

★ Unintended memorization

- Memorize out-of-distribution data

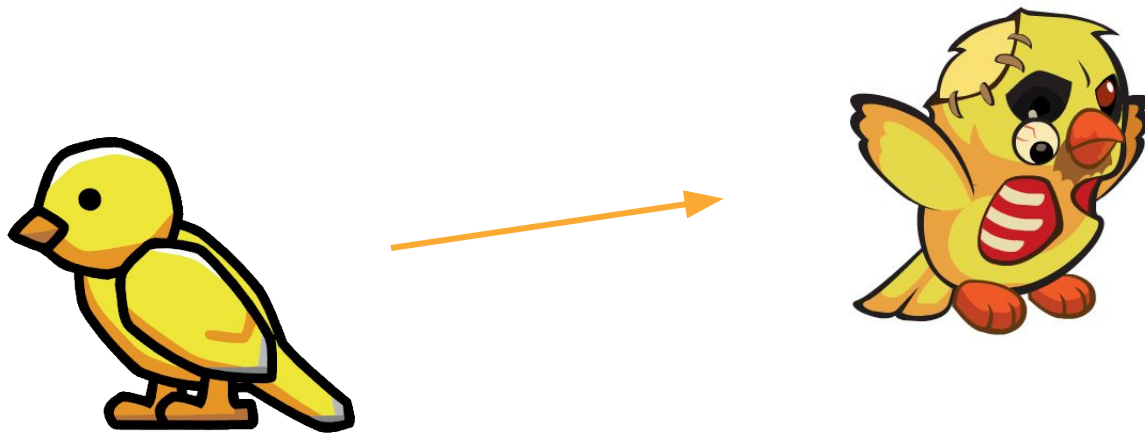
★ Secrets

- Data independent from data distribution to be learned

Canary: “The random number is * * * * * * * * * *”

Security Inspiration: Stack Canaries

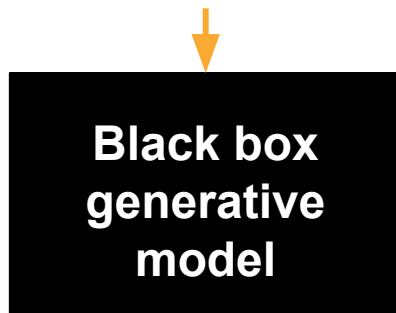
- ★ Randomly generated values inserted into the memory stack
- ★ If they are overwritten,
 - Then we detect an illegal overwriting of the stack



Out-of-Distribution Canary

- ★ We add phrases with randomly generated values to the training dataset for our NLP generative model

“The random number is _ _ _ _ _”



“The random number is * * * * *”

Reveals out-of-distribution canary!

Definition: Log-Perplexity

A measure of how perplexed/confused the model is to see this data during test time

Probability that we see this sequence given the trained model

$$\begin{aligned} P_{X_\theta}(x_1 \dots x_n) &= -\log_2 \mathbf{Pr}(x_1 \dots x_n | f_\theta) \\ &= \sum_{i=1}^n \left(-\log_2 \mathbf{Pr}(x_i | f_\theta(x_1 \dots x_{i-1})) \right) \end{aligned}$$

Definition: Rank

The lower the perplexity, the smaller the rank number

$$\mathbf{rank}_{\theta}(s[r]) = |\{r' \in \mathcal{R} : P_{\mathbf{x}_{\theta}}(s[r']) \leq P_{\mathbf{x}_{\theta}}(s[r])\}|$$

Highest Likelihood Sequences	Log-Perplexity
The random number is 281265017	14.63
The random number is 281265117	18.56
The random number is 281265011	19.01
The random number is 286265117	20.65
The random number is 528126501	20.88
The random number is 281266511	20.99
The random number is 287265017	20.99
The random number is 281265111	21.16
The random number is 281265010	21.36



Decreasing
rank number

Definition: Exposure

Size of space of randomness



$$\mathbf{exposure}_{\theta}(s[r]) = \log_2 |\mathcal{R}| - \log_2 \mathbf{rank}_{\theta}(s[r])$$



“Discretized” order of perplexity

Reminder: The lower the perplexity, the smaller the rank number

Exposure Hypothesis

If the canary is in the training data, we should see a lower rank value which means a high exposure value

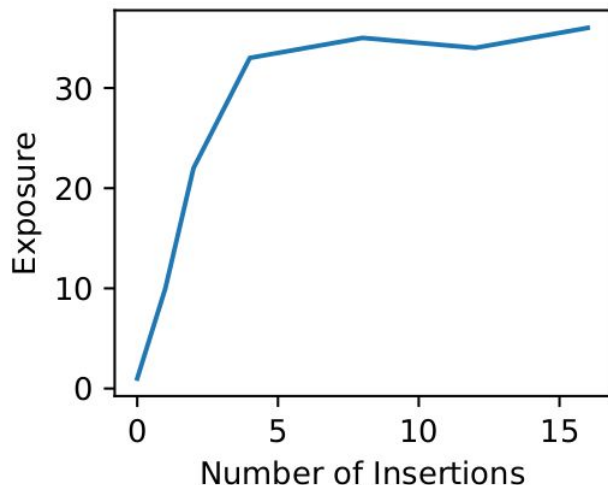
$$\text{exposure}_{\theta}(s[r]) = \log_2 |\mathcal{R}| - \log_2 \text{rank}_{\theta}(s[r])$$

Experiment: Neural Machine Translation

★ English-Vietnamese Recurrent Neural Network Model

★ Canary:

(“The random number is ”, Vietnamese +)



★ 4+ insertions of canary is fully memorized

Q: Isn't this just overfitting?

A: Yes, but not according to the typical definition.

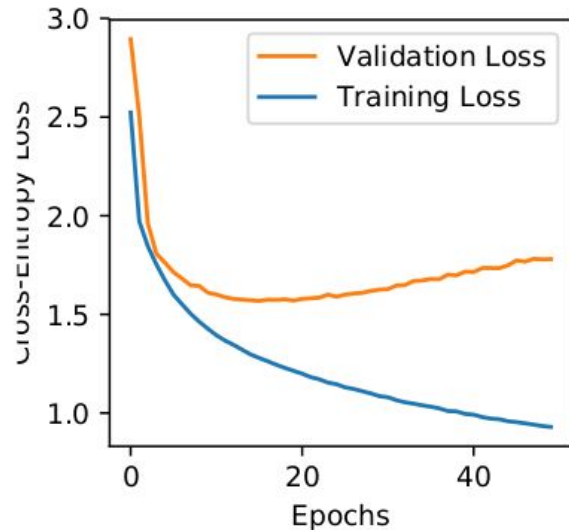
Definitions: Overfit vs Overtrain

★ Overfit

- Memorizes training data
- Does not generalize well to data within the same distribution

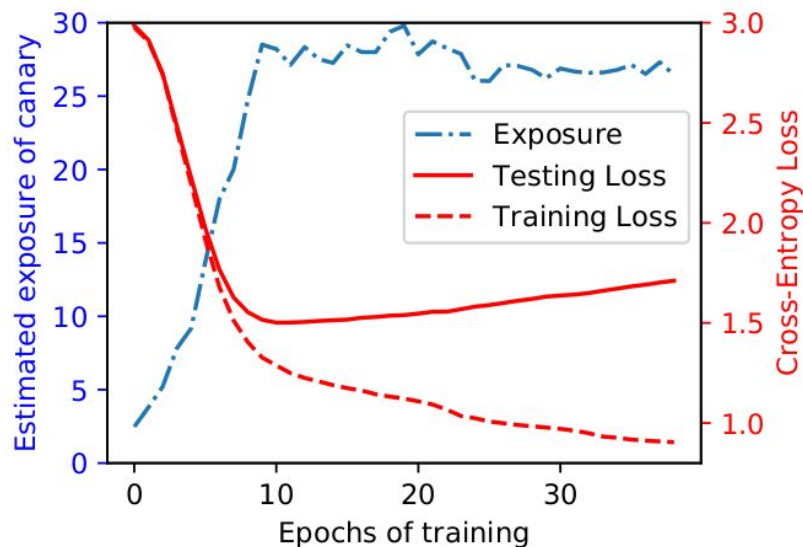
Authors clarify that their results are not due to overtraining

★ Overtrain



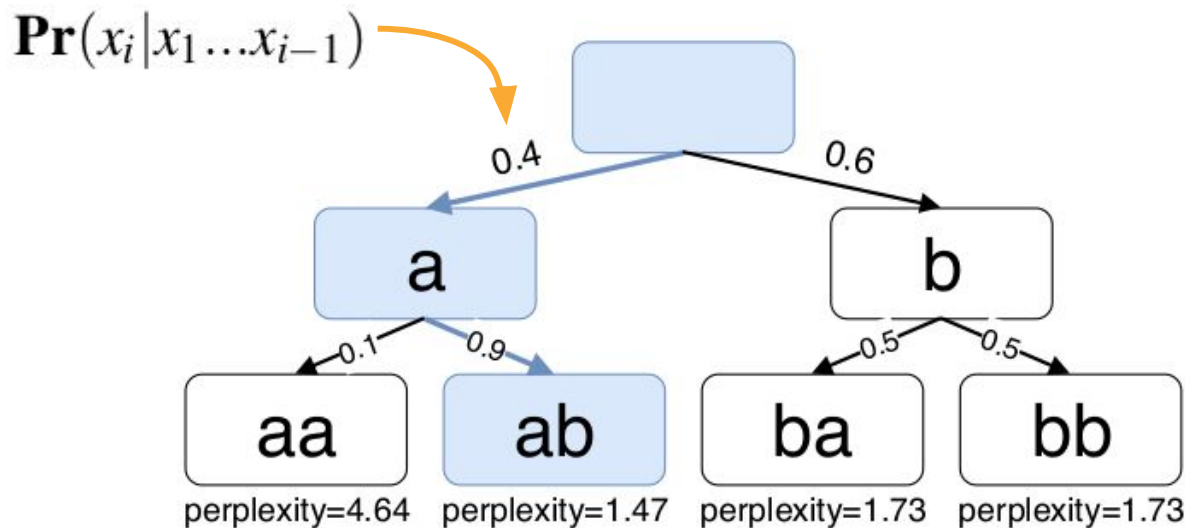
Why is it Not Due to Overtraining?

Model: Character-Level, Data: 5% of Penn Tree Bank (PTB) dataset



When the model starts overtraining, the exposure stays the same

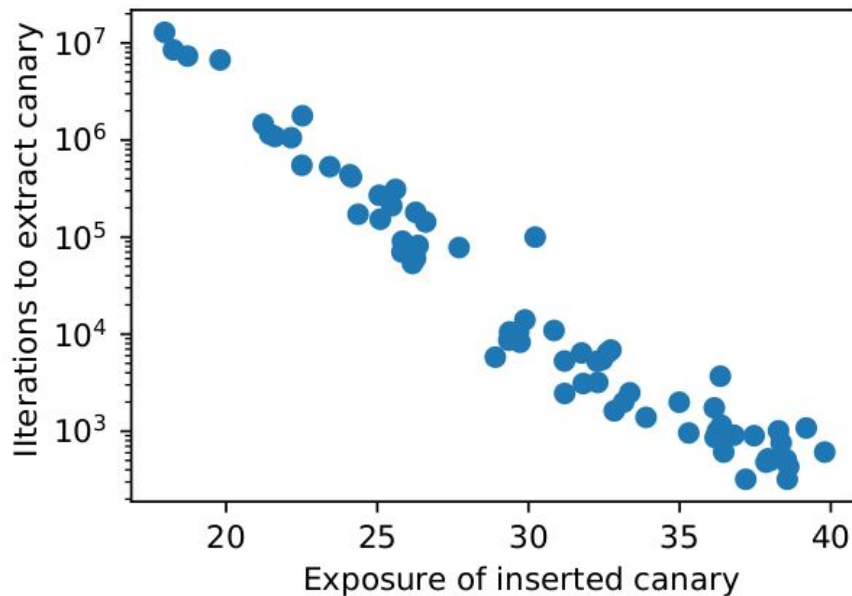
Extraction Algorithm: Shortest Path Search



Reminder: $\sum -\log_2 \Pr(x_i | f_\theta(x_1 \dots x_{i-1})) = \text{Px}_\theta(x_1 \dots x_n)$ Perplexity

Effectiveness of Extraction

- ★ Brute-force search
 - 10^9 attempts
- ★ Extraction algorithm
 - When exposure > 30 :
 10^4 attempts
 - 100,000× reduction

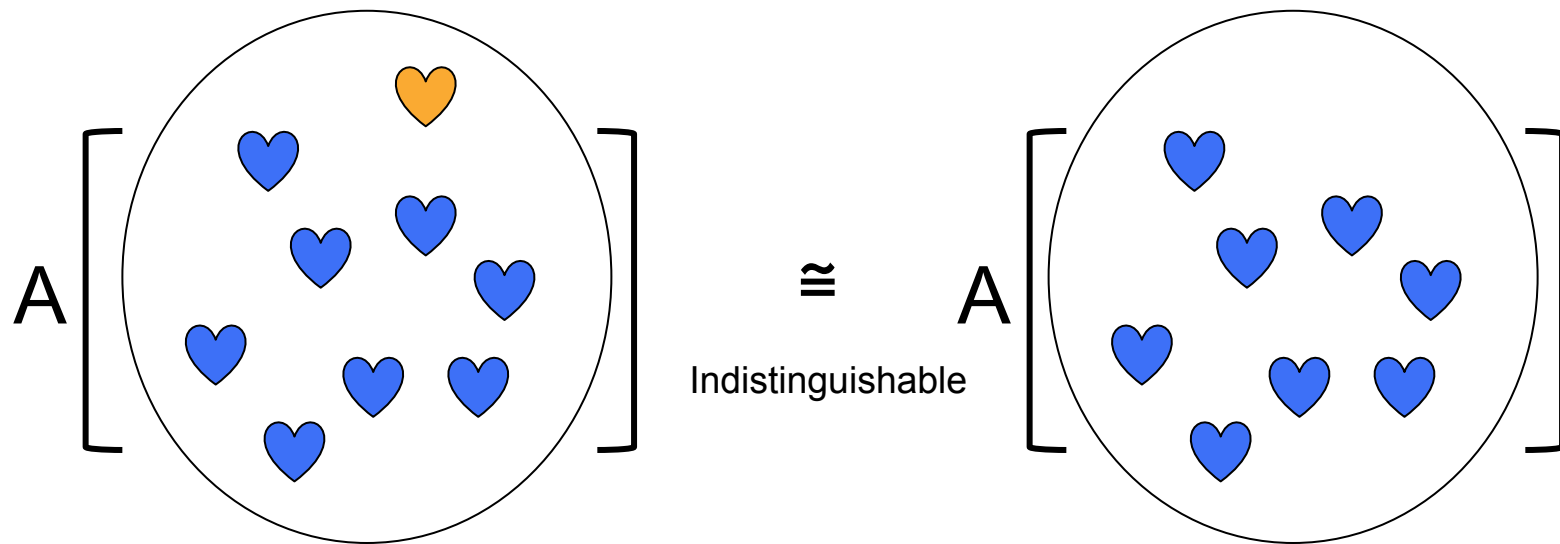


Experiment: Enron Email Dataset

User	Secret Type	Exposure	Extracted?
A	CCN	52	✓
B	SSN	13	
C	SSN	16	
	SSN	10	
	SSN	22	
D	SSN	32	✓
F	SSN	13	
G	CCN	36	
	CCN	29	
	CCN	48	✓

Took a few hours to extract the CCN/SSN numbers from the model

Definition: Differential Privacy



= user training data



= any specific user's data

Definition: Differential Privacy

Definition 5 *A randomized algorithm $\mathcal{A}$ operating on a dataset $\mathcal{D}$ is (ϵ, δ) -differentially private if*

$$\Pr[\mathcal{A}(\mathcal{D}) \in S] \leq \exp(\epsilon) \cdot \Pr[\mathcal{A}(\mathcal{D}') \in S] + \delta$$

for any set S of possible outputs of $\mathcal{A}$, and any two data sets $\mathcal{D}, \mathcal{D}'$ that differ in at most one element.

Strongest privacy guarantees: $\epsilon = 1$ and $\delta < |\mathcal{X}|^{-1}$

Defense: Differential Privacy

	Optimizer	ϵ	Test Loss	Estimated Exposure	Extraction Possible?
With DP	RMSProp	0.65	1.69	1.1	
	RMSProp	1.21	1.59	2.3	
	RMSProp	5.26	1.41	1.8	
	RMSProp	89	1.34	2.1	
	RMSProp	2×10^8	1.32	3.2	
	RMSProp	1×10^9	1.26	2.8	
	SGD	∞	2.11	3.6	
No DP	SGD	N/A	1.86	9.5	
	RMSProp	N/A	1.17	31.0	✓

Strategy: Clip the per-example gradient to a max norm and add Gaussian noise

Deep Models Under the GAN (2017)

Deep Models Under the GAN: Information Leakage from Collaborative Deep Learning

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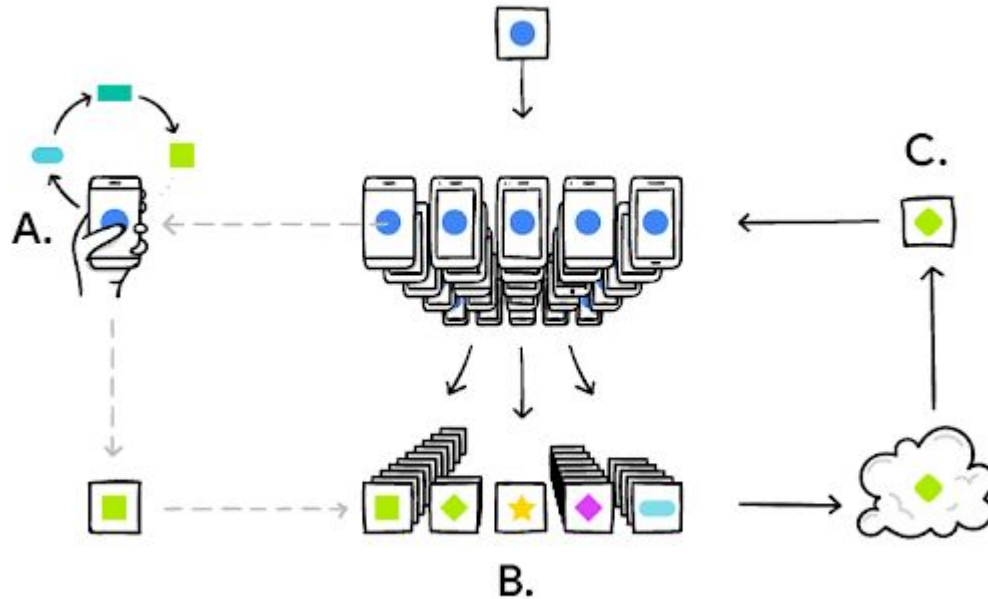
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Paper Overview

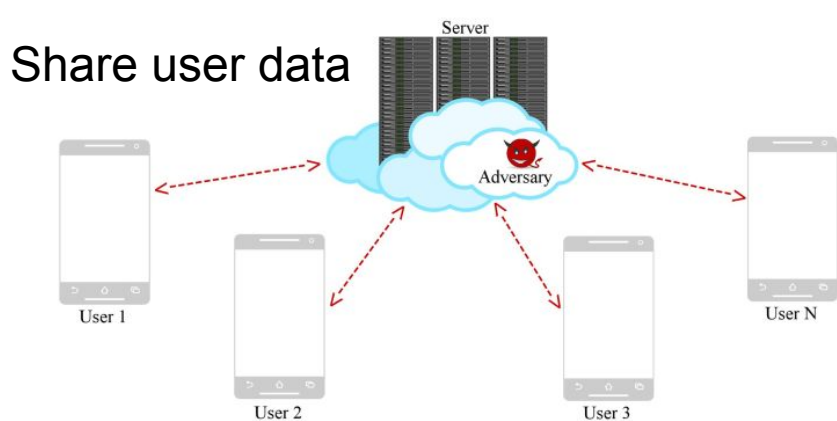
- ★ New setting: federated learning
- ★ How does this new GAN attack compare with Model Inversion Attack (MIA)?
 - GAN is more powerful due to its “online” nature
 - Naturally leads to adaptive attacks
- ★ What happens with we add Differential Privacy (DP)?
 - Adaptive GAN attack still succeeds

Setting: Federated (or Collaborative) Learning

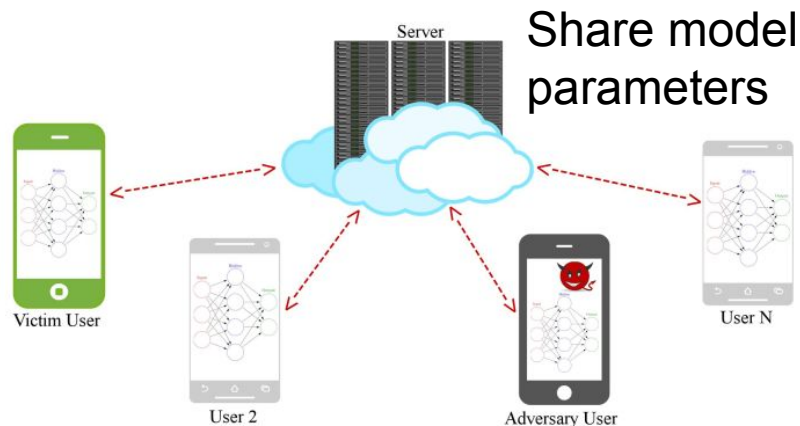


Centralized vs Collaborative Adversary

Previously only the server could be adversarial and break privacy, but now each user has the power of an adversary



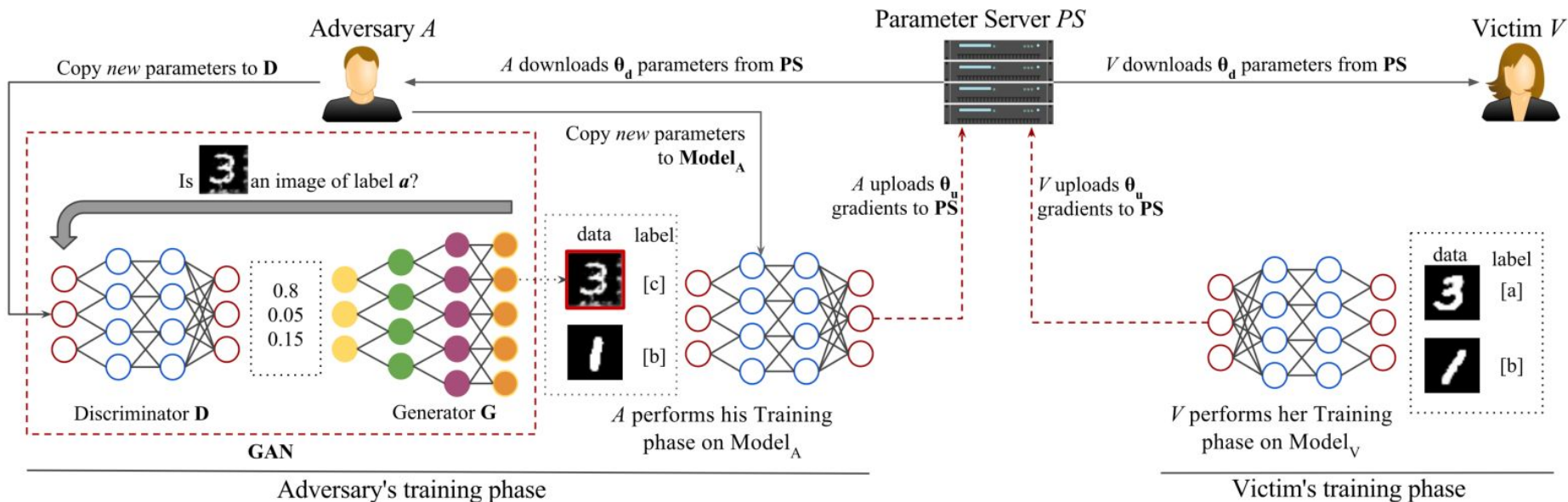
(a) Centralized Learning



(b) Collaborative Learning

GAN Attack on Collaborative Learning

Motivation: GANs devised for density estimation, so can learn data distribution from the model output without direct access to the data

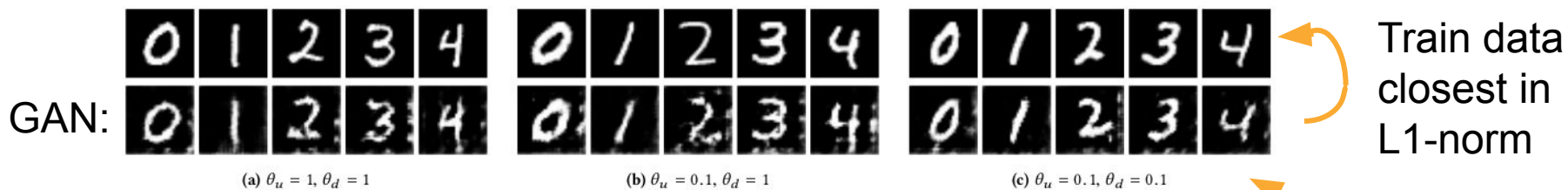


Model Inversion vs GAN

- ★ Adversary is adaptively malicious
 - Curious about a class of data they do not own
- ★ Model Inversion Attack (MIA)
 - Against final trained model
 - Effective against multi-layer perceptron (MLP), but not against convolutional neural networks (CNN)
- ★ GAN Attack (DCGAN)
 - Uses real-time access to influence training phase
 - Effective against CNN (see experiment on right)

Actual Image	MIA	DCGAN
		
		
		
		
		
		
		
		
		
		

Experiments of GAN Attack: MNIST data

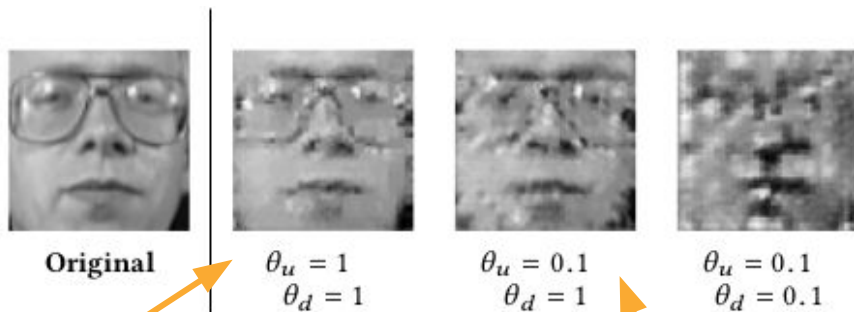


(a) users upload and download the entire model

(b) users download the full model, upload 10% of the parameters in each epoch

(c) upload and download is 10%

Experiments of GAN Attack: AT&T data



AT&T dataset is smaller and accuracy degrades when user uploads are small

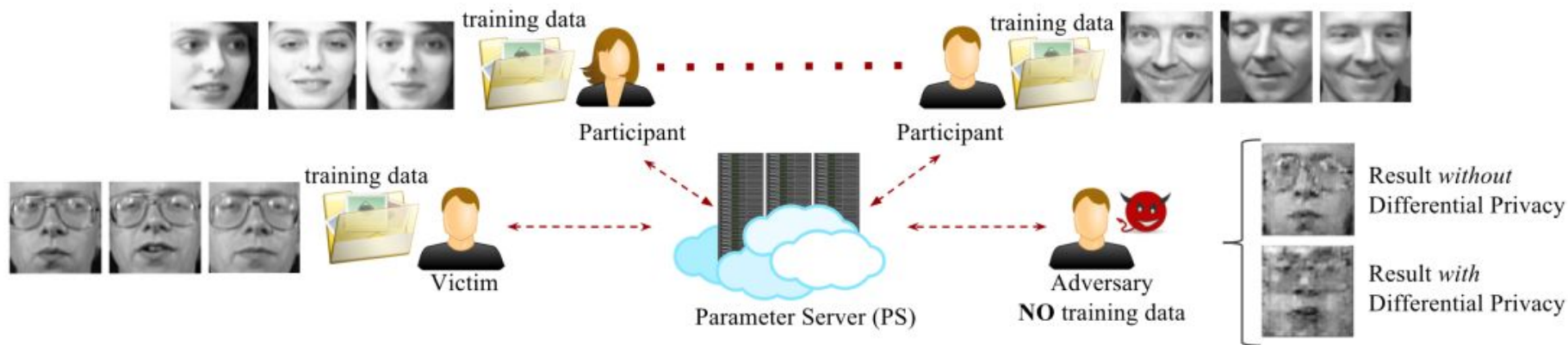
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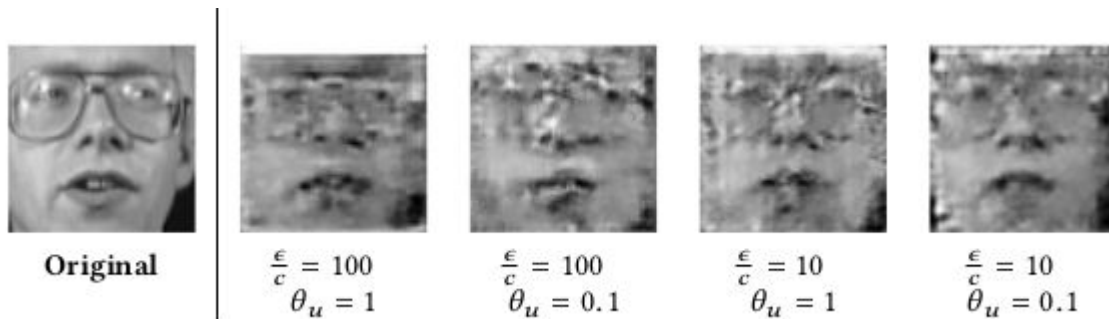
(c) upload and download is 10%

Bypasses “Record-Level” Differential Privacy

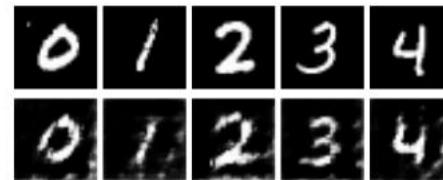
- ★ 41 users (1 malicious adversary)
- ★ GAN attack able to reconstruct victim’s private data even with differential privacy (DP) enabled



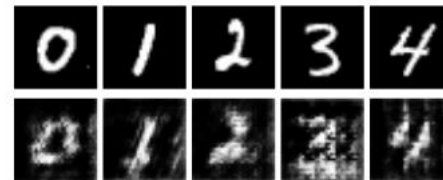
Experiments with DP



Less effective, but still reconstructable



(a) $\frac{\epsilon}{c} = 100$, $\theta_u = 1$, $\theta_d = 1$



(b) $\frac{\epsilon}{c} = 100$, $\theta_u = 0.1$, $\theta_d = 1$



(c) $\frac{\epsilon}{c} = 10$, $\theta_u = 1$, $\theta_d = 1$



(d) $\frac{\epsilon}{c} = 10$, $\theta_u = 0.1$, $\theta_d = 1$

Why Doesn't "Record-Level" DP Suffice?

- ★ GAN attack is effective even using differential privacy
 - Authors say that the attack relies only on the accuracy of the discriminative model of the GAN and not its actual gradient values

Summary of Strategy

★ The Secret Sharer (2019)

- Demonstrated privacy attack on NLP generative models
- Designed metric to measure exposure of out-of-distribution data
- Analyzed techniques to defend against this attack
- Concluded differential privacy was secure

★ Deep Models Under the GAN (2017)

- Demonstrate GAN attack to reconstruct data in FL setting
- Apply suggested defenses: “record-level” DP
- Demonstrate attack against FL + “record-level” DP

Future Directions

- ★ How to prevent memorization of out-of-distribution data in a way that doesn't sacrifice accuracy of the model?
- ★ How to build collaborative learning system using user or device-level differential privacy?
- ★ Is the GAN attack reproducible in NLP setting?